

Quantum Simulation of Real Materials: Electronic Band Structure Analysis

Problem Statement & Project Pipeline

1st April 2026

1 Problem Statement

Real materials contain electrons whose quantum mechanical interactions determine all electronic properties such as conductivity, band gaps, and magnetism. While Density Functional Theory (DFT) is a standard approach, the exact solution is computationally intractable on classical computers for strongly correlated systems where DFT fails.

Quantum computers offer a natural representation of fermionic quantum states; however, current Noisy Intermediate-Scale Quantum (NISQ) hardware is too noisy for large systems. The goal of this project is to build a full pipeline—from crystal structure to quantum Hamiltonian to quantum circuit—to extract the electronic band structure for a real material.

2 Methodology and Target Materials

We shall use **Graphene** as the primary target due to the availability of analytical, DFT, and quantum simulation references. This allows for a clean comparison of how different methods (DFT, Wannier, VQE, ZNE) succeed or fail at reproducing the Dirac-cone physics. As a stretch goal, we will target the **MgO vacancy**, where quantum simulation adds genuine physical value beyond standard DFT for defect spectroscopy.

3 Project Pipeline and Key Steps

Step 1: Literature Review + Classical Reference (May – June)

Task: Equip yourself with the necessary technical background in condensed matter physics and quantum mechanics, with particular emphasis on electronic structure methods. Once this foundation is established, perform first-principles calculations on graphene using Quantum ESPRESSO by solving the Kohn–Sham equations within Density Functional Theory within different approximations. This involves converging the self-consistent charge density (SCF cycle) and subsequently computing the electronic band structure along high-symmetry paths in the Brillouin zone.

Outcome: A first-principles (ab initio) reference for graphene’s electronic structure, capturing the full periodic crystal potential and all relevant orbital contributions, obtained via quantum mechanical calculations executed on classical computational hardware.

Step 2: Model Downfolding (June)

Task: Use Wannier90 to downfold the DFT-computed band structure of graphene into a minimal tight-binding Hamiltonian defined in a reduced Hilbert space. Specifically, construct two s -like Wannier functions (one per carbon atom in the unit cell, derived from the p_z orbital manifold) and express the system as an effective lattice Hamiltonian with a limited set of hopping parameters.

This step maps the full Kohn-Sham Hamiltonian onto a low-energy subspace suitable for encoding on quantum hardware, significantly reducing the number of degrees of freedom while preserving the essential physics.

Outcome: A compact tight-binding Hamiltonian that faithfully reproduces the low-energy π -band structure of graphene, particularly the Dirac cone dispersion which is a signature feature in the electronic structure of graphene. It is directly amenable to implementation on quantum computing platforms (e.g., via second-quantized fermionic encodings), avoiding the prohibitive cost of simulating the full ab initio Hamiltonian..

Step 3: Hamiltonian Construction (July)

Task: Starting from the Wannierized tight-binding model, construct the second-quantized lattice Hamiltonian using tools such as OpenFermion or PySCF. Express the Hamiltonian in terms of fermionic creation and annihilation operators, including hopping terms between Wannier orbitals.

Apply a fermion-to-qubit transformation such as the Jordan-Wigner transformation to map the fermionic Hamiltonian onto Pauli operators acting on qubits..

Outcome: A qubit-encoded Hamiltonian expressed as a weighted sum of Pauli strings. The number of qubits scales with the number of spin-orbitals (typically greater than or equal to 2 per Wannier orbital), yielding a compact representation suitable for quantum simulation frameworks such as Qiskit.

Step 4: VQE Implementation (August – September)

Task: For each k -point along the $\Gamma \rightarrow K \rightarrow M$ path in the Brillouin zone, construct the corresponding k -space tight-binding Hamiltonian and solve for its lowest eigenvalue using the Variational Quantum Eigensolver (VQE) within Qiskit.

Given that graphene's π -band model is fundamentally non-interacting, use a hardware-efficient ansatz or a minimal parameterized circuit rather than a full Unitary Coupled Cluster (unless electron-electron interactions are explicitly introduced). Optimize circuit parameters to obtain the ground-state energy at each k -point.

Outcome: Ground-state energy eigenvalues $E(k)$ for each sampled k -point along the high-symmetry path..

Step 5: Ideal Simulation & Verification (October)

Task: Assemble the computed eigenvalues $E(k)$ across the $\Gamma \rightarrow K \rightarrow M$ path to reconstruct the band structure. Compare the resulting dispersion against analytical tight-binding predictions and the original DFT results.

Outcome: SRecovery of the characteristic graphene band structure, including the linear Dirac cone at the K point, demonstrating that the downfolded Hamiltonian and its quantum encoding faithfully reproduce the low-energy physics under ideal (noise-free) quantum simulation conditions.

Step 6: Hardware Execution & Noise Analysis (October – November)

Task: Execute the VQE workflow for selected k-points on real quantum hardware using IBM Quantum. Measure the expectation value of the qubit Hamiltonian to obtain energy estimates $E(k)$ in the presence of device noise (gate errors, decoherence, readout errors).

Apply error mitigation techniques such as Zero-Noise Extrapolation (ZNE) by systematically amplifying circuit noise and extrapolating expectation values to the zero-noise limit..

Outcome: A set of noisy and error-mitigated energy estimates $E(k)$, enabling direct comparison between ideal simulator results, hardware-executed results, and ZNE-corrected values

Overall Outcome till now: A quantitative benchmark of quantum band structure simulation on NISQ hardware. Noise-induced errors manifest as distortions in the dispersion, particularly near the K point, leading to apparent smearing or lifting of the Dirac-point degeneracy. The effectiveness of Zero-Noise Extrapolation is assessed via recovery of the linear Dirac dispersion relative to ideal and Density Functional Theory results.

Metrics such as Dirac cone slope deviation, apparent gap at K, and overall energy error provide a compact, reusable benchmark for evaluating quantum hardware and error mitigation performance.

4 Stretch Goal: MgO Oxygen Vacancy

We will apply the DFT embedding + VQE/VQD (Variational Quantum Deflation) scheme to the MgO oxygen vacancy . The goal is to compute the energy difference between ground and excited states and compare the predicted emission energy to the experimental value of ≈ 2.3 eV .

5 Result of the Project

- This project provides a self-contained, end-to-end pipeline to obtain and compare the band structure of graphene through several distinct methods: classical DFT, Wannier-tight-binding, ideal quantum simulation (VQE on a simulator) , noisy quantum hardware simulation (on real IBM Quantum devices) , and zero-noise extrapolation (ZNE) .
- By overlaying these band structures and focusing on the Dirac cone at the K point , the project systematically quantifies how NISQ-scale hardware noise distorts the band dispersion, measures the emergence of fake gaps, and evaluates how much ZNE can recover the ideal band shape. These results constitute a clean, physics-rich benchmark that future researchers can use to test the performance of their own quantum hardware, noise-mitigation techniques, and band-structure algorithms on a well-understood solid-state system.
- Once this graphene-based benchmark is established, the same pipeline is extended to the MgO oxygen vacancy . Here, instead of reconstructing a full band structure, the project

focuses on the defect-spectroscopy observable : the emission energy of the vacancy, which is experimentally known to lie around 2.3 eV .

- Using DFT embedding , the environment of the MgO host is treated with classical DFT, while the low-energy defect orbitals around the vacancy are mapped to qubits and treated with VQE for the ground state and VQD for the excited state .
- The resulting energy difference is compared to both standard DFT/TDDFT predictions and the experimental value, directly assessing how much quantum simulation improves the description of this strongly correlated defect transition beyond what DFT alone can achieve.
- In this way, the MgO part extends the benchmarking philosophy from band-structure physics to defect-spectroscopy physics : - For graphene, the benchmark is “how cleanly can current hardware reproduce a known Dirac-cone band structure?” - For MgO, the benchmark becomes “how reliably can embedded quantum simulation reproduce a known defect emission energy, and how much does it improve on DFT?”
- Taken together, this project serves as a dual-purpose benchmark suite—one for band-structure simulations in solids and another for defect-spectroscopy simulations—that future researchers can use as a reference when evaluating new quantum-materials algorithms, error-mitigation schemes, and hardware platforms.

6 Technical Tools and Softwares Required

- Quantum Espresso (Open Source Software)
- Wannier90 (Open Source Software)
- Open Fermion (Python Library)
- PySCF (Python Library)
- Qiskit (Python Library)
- IBM Hardware (IBM gives 10 minute free use per 28 days for free)
- The softwares mentioned above are optimised for Linux use. Hence, Linux.
- Google One Storage Plan or External Hard Drive for large simulation files
- AI tools like Perplexity, Claude, etc.